CALCULUS · UNIT 8: DIFFERENTIAL EQUATIONS

Cold Case: Newton's Cooling

Use differential equations, slope fields, and Euler's method to solve a mystery with Newton's Law of Cooling.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A mug of cocoa was found on the desk of a missing museum key. The temperature of the cocoa was measured twice, and the room temperature is known. Newton's Law of Cooling says the cocoa cools at a rate proportional to the difference between its temperature and the room.

You will use differential equations to decide when the cocoa was poured and which suspect was in the room at that time.



Room

Constant 68°F

Cocoa

9:00 p.m.: 140°F

9:30 p.m.: 116°F

Learning Goals

- I can verify a solution of a differential equation.
- I can read and sketch a slope field.
- I can use an exponential model to solve for time.

Project Timeline

When	What you will do
Day 1	Rates of Cooling
Day 2	Slope Field
Day 3	Solve the Case
Final	Case File Report + reflection

Part 1: Rates of Cooling

Day 1 · Newton's Law: $\frac{dT}{dt} = -k(T - 68)$.

Evidence Cards

Room Temp	Reading 1	Reading 2
68°F all night	9:00 p.m. · 140°F	9:30 p.m. · 116°F

Cocoa Poured	Suspect A	Suspect B
Fresh cocoa is 185°F	In the room	In the room
	8:15–8:35 p.m.	8:35–8:55 p.m.

1. Is $\frac{dT}{dt}$ positive or negative when the cocoa is 140°F? What does that mean?

2. Where is the cocoa cooling fastest: at 140°F or 100°F? Why?

3. Verify that T = 68 + 72e^(−kt) satisfies $\frac{dT}{dt} = -k(T - 68)$.

Starter: *Differentiate T, then compare with −k(T − 68).*

4. What happens to T as t becomes very large?

Part 2: Slope Field

Day 2 · Use k = 0.02 per minute: $\frac{dT}{dt} = -0.02(T - 68)$.

5. Find the slope at T = 140, T = 104, T = 68, and T = 50.

6. Sketch the slope field for 0 ≤ t ≤ 60 and 40 ≤ T ≤ 160. Describe the pattern.

7. Sketch the solution through (0, 140).

8. Explain why the line T = 68 is a solution.

Part 3: Solve the Case

Day 3 · With t = 0 at 9:00 p.m.: T(t) = 68 + 72e^(−kt) and T(30) = 116.

9. Find k.

10. When was the cocoa 185°F? Solve 68 + 72e^(−kt) = 185 (t will be negative).

11. Which suspect was in the room when the cocoa was poured?

Final Product: Case File Report

Your final product must include:

- Verified model
- Slope field with a solution curve
- Value of k
- Time the cocoa was poured
- Case file naming the suspect

Reflection

R1. Why does the cocoa cool more slowly over time?

R2. Why was t negative when solving for 185°F?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Differential equations are analyzed and solved correctly and used to reach a justified verdict. Goes beyond the minimum with a creative, realistic choice.	Differential equations are analyzed and solved correctly and used to reach a justified verdict.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Case file includes the model, slope field, calculations, uncertainty, and verdict. Polished, organized, and visually engaging.	Case file includes the model, slope field, calculations, uncertainty, and verdict. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.